

Many Steps of Evaluation

Given a relation $\rightarrow$, we define a new relation $\rightarrow^$ by:*

$E \rightarrow^ E'$ holds if and only if either $E = E'$ (so no steps of evaluation are needed to get from E to E') or there is a finite sequence*

$$E \rightarrow E_1 \rightarrow E_2 \dots \rightarrow E_k \rightarrow E'.$$

This relation $\rightarrow^$ is called the reflexive transitive closure of $\rightarrow$.*

For our expressions, we say that number n is the final answer of E if $E \rightarrow^ n$.*